Streaming Platform Relational Database Design and Analysis

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1. Introduction

In this project, a relational database is designed and implemented for a hypothetical streaming service. Essentially, the goal was to replicate the way an actual streaming service might store data on users, content, and user engagement with content. This database was built using a structured relational database methodology and was populated with synthetic data to respect ethical standards.

The database is not just meant to store information but also to allow for insightful analyses and visualizations. Care was taken in the database design, especially in terms of referential integrity, synthetic data creation, and ethical considerations related to user information.

2. Database Structure and Design Rationale

A database with six tables has been designed. Each of these tables represents a different part of the world of a streaming service. The interaction tables are larger than the content and user tables because user actions are much more frequent than content creation.

- **Users (1100 records)** store customer information and subscription type.
- **Movies (300 records)** contain content details such as genre and duration.
- **Watch History (2000 rows)** records viewing activity and enables time-based analysis.
- **Ratings (1200 records)** capture user feedback on content.
- **Payments (1200 records)** track subscription transactions.
- **Engagement (1500 records)** measures behavioral intensity (e.g., minutes watched).

This schema uses relational modeling and is in normal form up to 3NF to minimize redundancy and avoid update anomalies. Each table has primary keys to uniquely identify all the records in the table. Users are at the center with one-to-many relationships to other behavior tables such as Watch History, Payments, and Engagement. However, there is a many-to-many relationship between Users and Movies, and this is done nicely with join tables (Ratings and Watch History) to show how users interact with movies.

Composite keys are used in interaction tables such as Ratings (User-ID, Movie-ID) and Watch History (User-ID, Movie-ID, Watch-Date) to prevent illogical duplicate entries but at the same time allow a user to interact with a movie repeatedly over time. Overall this schema design is faithful to relational database modeling and 3NF normalization.

3. Data Generation and Realism

The data was created artificially using randomized logic to mimic what it would look like if it were created by a real streaming service. I have also added missing data in some cases to mimic what it would look like in real data. For instance, 2 to 5 percent of demographic data, 10 percent of behavioral data, and 25 percent of financial data are missing in our dataset. The tables that store interaction data, like Watch History, Ratings, and Engagement, have

significantly more data than the User or Movie tables to mimic the high volume of user-content interaction you would expect in a streaming service.

4. Data Types and Integrity

The schema includes a variety of data types to accommodate different information:

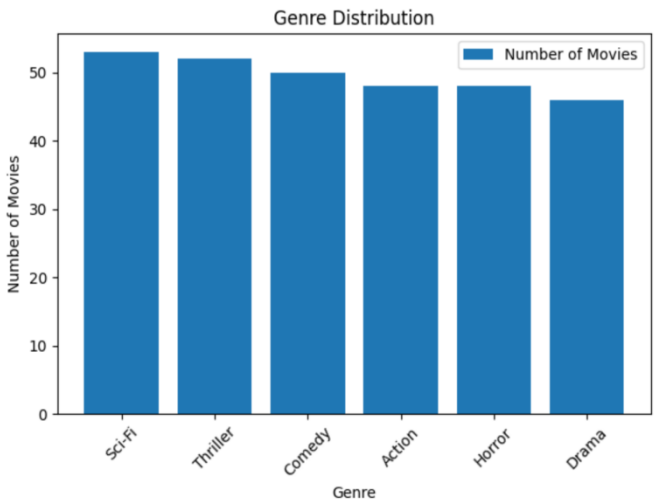
- INTEGER for unique identifiers
- TEXT for descriptive and categorical attributes
- REAL for financial and engagement metrics
- Date-based fields for temporal tracking

We have employed attribute-level constraints, e.g., NOT NULL, UNIQUE, and rating ranges, to ensure logical consistency and domain integrity. Data distribution is intentionally uneven and not artificially uniform for any attribute, e.g., engagement levels, subscription levels, and ratings.

5. Analytical Visualizations

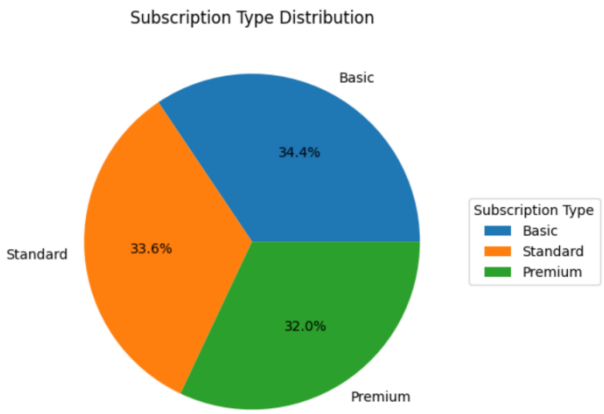
To showcase what is possible with the database, three different visualizations were created.

5.1 Genre Distribution (Bar Plot)



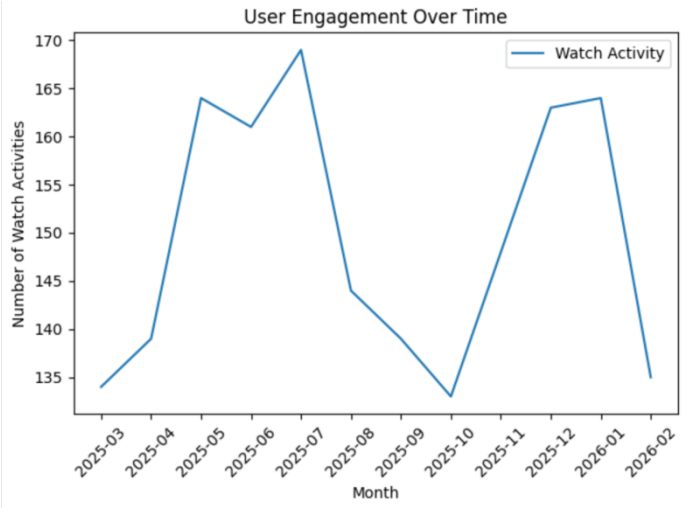
The breakdown of genres illustrates the construction of the platform's content. Divergence among genres is due to intentional diversification, aiding in strategic analysis of the content. Such analysis is essential in real life for determining what is licensed and who is targeted.

5.2 Subscription Type Distribution (Pie Plot)



The break-down of subscriptions illustrates user distribution across different pricing levels. Unequal proportions mirror market diversity and help assess revenue distribution across different levels. Such analysis is crucial for pricing optimization and business modeling of streaming platforms.

5.3 Engagement over Times (Line Plot)



The engagement trend illustrates user engagement over a period of time and is essential for analysis of patterns of behavior. Normalization was applied where necessary for accurate analysis over an incomplete timeframe. Such analysis is essential for demand forecasting and infrastructure planning.

6. Ethical and Privacy Considerations

However, even with artificial information, actual platforms store sensitive information regarding users' personality and behavior. Ethical considerations for this type of data include ensuring the data is kept safe, who can access it, and ensuring that data privacy regulations such as the GDPR are met. The GDPR emphasizes data minimization, obtaining users' consent, and protecting personal data. This project intentionally avoids actual user information and illustrates responsible database design. The design approach is similar to responsible privacy-focused data system design best practices.

7. Limitations

Since the information is artificially created, it is possible that some of the behaviors might not fully represent actual user interactions. However, it accurately represents professional standards for a database design and its integrity.

8. Conclusion

This project presents an approach for the design of a relational database for the streaming service. Through normalization, referential integrity, realistic data, and analytical visualization, it effectively represents user interactions, content management, and transactional user behavior. The database design is similar to professional relational database design principles and is suitable for analytical and ethically grounded analytical exploration.

The Python code used to generate the database is available in the submitted **jupyter notebook file** and can also be accessed via the **GitHub repository**:
https://github.com/Arman-025/Data-Mining-and-Discovery-Database-Code-